Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Billakanti Siri
Roll Number	2520030601
Branch	CSE
Course Outcome	CO5 – LSD Radix Sort for Large Numeric Keys

Case Study Topic: Aadhaar UID De-duplication using LSD Radix Sort

Work Done Summary:

In this case study, we implemented LSD (Least Significant Digit) Radix Sort to sort Aadhaar UID numbers efficiently. Since all UID numbers contain fixed-length numeric keys, radix sort provides faster performance compared to traditional comparison-based sorting algorithms. Stable counting sort was used internally during each digit pass to maintain correct ordering.

Implementation Details:

1. Implemented LSD Radix Sort using Java.
2. Applied digit-wise sorting from least significant digit to most significant digit.
3. Used stable Counting Sort during every pass.
4. Implemented bucket-based digit grouping for digits 0–9.
5. Performed sorting in three passes:

Result: Screenshots/output

```
siribillakanti@siris-MacBook-Air VSC DSA II % javac assignment_CO5.
siribillakanti@siris-MacBook-Air VSC DSA II % java assignment_CO5
Original Array:
[473, 152, 681, 247, 539, 826, 715, 304]

After Pass 1 : [681, 152, 473, 304, 715, 826, 247, 539]

After Pass 2 : [304, 715, 826, 539, 247, 152, 473, 681]

After Pass 3 : [152, 247, 304, 473, 539, 681, 715, 826]

Final Sorted Array:
[152, 247, 304, 473, 539, 681, 715, 826]
siribillakanti@siris-MacBook-Air VSC DSA II %
```

GitHub Link:
https://github.com/siribillakanti/DSA2_ASSIGNMENTS/blob/main/assignment_CO5.java